Algebraic structures

Binary operation :- Let 'S' be a non empty set. If $f : S \times S \to S$ is a mapping, then 'f' is called a binary operation or binary composition in 'S'. The symbols $+, \cdot, *, \circ, \oplus$ etc are used to denote binary operations

Ex: For $a, b \in S \Rightarrow a+b \in S \Rightarrow$ '+' is a binary operation

For $a, b \in S \Rightarrow a*b \in S \Rightarrow$ '*' is a binary operation.

This is said to be the closure property of the binary operation and the set 'S' is said to be closed with respect to the binary operation.

Algebraic structures :- A non empty set 'G' equipped with one or more binary operations is called an algebraic structure or an algebraic system. If '*' is a binary operation on G then the algebraic structure is written as $(G, *)$.

Ex: $(N, +)$, $(\Theta, -)$, $(R, +)$ etc are algebraic structures

Semi group :- An algebraic structure $(G, *)$ is called a semi group if the binary operation '*' is a binary operation Then $(G, *)$ is said to be a semi group if i) $a, b \in G \Rightarrow a*b \in G$ for all $a, b \in G$

ii) $a*(b*c) = (a*b)*c$ for all $a, b, c \in G$

i) property is called closure property

ii) property is called Associative property.

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Ex: The operation * is defined by $a * b = a$ for all $a, b \in G$ show that $(G, *)$ is a semi group.

Sol: Let $a, b \in G \Rightarrow a * b = a \in G$

$\therefore$ '*' is a binary operation in G

i) closure property : Let $a, b \in G$ then $a * b = a \in G$

ii) Associative property :- Let $a, b, c \in G$

$a * (b * c) = a * b = a$

$(a * b) * c = a * c = a$

$\therefore a * (b * c) = (a * b) * c$

'*' is associative in G

Hence $(G, *)$ is a semi group.

Ex: The operation * is defined by $a * b = a + b - ab$ for all $a, b \in Z$ show that $(Z, *)$ is a semi group.

Sol: Let $a, b \in Z \Rightarrow a * b = a + b - ab \in Z$

$\therefore$ '*' is a binary operation in G

i) closure property :- Let $a, b \in G$ then $a * b = a + b - ab \in G$

ii) Associative property :- Let $a, b, c \in G$

$a * (b * c) = a * (b + c - bc)$

$= a + (b + c - bc) - a(b + c - bc)$

$= a + b + c - bc - ab - ac + abc$

$(a * b) * c = (a + b - ab) * c$

$= a + b - ab + c - (a + b - ab) \cdot c$

$= a + b + c - ab - ac - bc - abc$

$\therefore a * (b * c) = (a * b) * c$

Hence $(G, *)$ is a semi group.

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Ex: '$\mathbb{Q}$' is the set of rational numbers, $*$ is a binary operation defined on $\mathbb{Q}$ such that $a*b = a-b+ab$ for $a,b \in \mathbb{Q}$ Then $(\mathbb{Q}, *)$ is not a semi group.

Sol: Let $a,b \in \mathbb{Q} \Rightarrow a*b = a-b+ab \in \mathbb{Q}$

'$*$' is a binary operation in $\mathbb{Q}$.

i) closure property : Let $a,b \in \mathbb{Q} \Rightarrow a*b = a-b+ab \in \mathbb{Q}$

ii) Associative property : Let $a,b,c \in \mathbb{Q}$

$$a*(b*c) = a*(b-c+bc)$$
$$= a-(b-c+bc)+a(b-c+bc)$$
$$= a-b+c-bc+ab-ac+abc$$
$$(a*b)*c = (a-b+ab)*c$$
$$= (a-b+ab)-c+(a-b+ab)\cdot c$$
$$= a-b-c+ab+ac-bc+abc$$

$\therefore a*(b*c) \neq (a*b)*c$

Hence $(\mathbb{Q}, *)$ is not a semi group.

Ex: Let $(A,*)$ be a semi group. Show that for a,b,c in A if $a*c = c*a$ and $b*c = c*b$ then $(a*b)*c = c*(a*b)$.

Sol: Given $(A,*)$ is semi group

$a*c = c*a$ and $b*c = c*b$ for $a,b,c \in A$

Now $(a*b)*c = a*(b*c)$ $\quad \langle \because$ Associative

$= a*(c*b)$ $\quad \langle \because c*b = b*c$

$= (a*c)*b$ $\quad \langle \because$ Associative

$= (c*a)*b$ $\quad \langle \because a*c = c*a$

$= c*(a*b)$ $\quad \langle \because$ Associative

$\therefore a*(b*c) = c*(a*b)$

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Ex: Shows that the binary operation * defined on $(R, *)$ where $x * y = \max(x, y)$ is a semi group.

Sol: Let $x, y \in R \Rightarrow x * y = \max(x, y) \in R$

'*' is a binary operation in R

i) closure property : Let $x, y \in R \Rightarrow x * y = \max(x, y) \in R$

ii) Associative property :- Let $x, y, z \in R$

$$(x * y) * z = \max(x, y) * z$$

$$= \max(\max(x, y), z) = \max(x, y, z)$$

$$x * (y * z) = x * \max(y, z)$$

$$= \max(x, \max(y, z)) = \max(x, y, z)$$

$$\therefore x * (y * z) = (x * y) * z$$

Hence $(R, *)$ is a semi group.

Ex :- Shows that the binary operation * defined on $(R, *)$ where $x * y = x^y$ is not a semi group.

Sol: Let $x, y \in R \Rightarrow x * y = x^y \in R$

'*' is a binary operation in R

i) closure property :- Let $x, y \in R \Rightarrow x * y = x^y \in R$

ii) Associative property : Let $a, b, c \in R$

$$(x * y) * z = x^y * z$$

$$= (x^y)^z = x^{yz}$$

$$x * (y * z) = x * y^z$$

$$= x^{y^z}$$

$$\therefore (x * y) * z \neq x * (y * z)$$

Hence $(R, *)$ is not a semi group.

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Homomorphism of semi-group :-

Let $(S, *)$ and (T, o) be any two semi-groups. A mapping $f: S \to T$ such that for any two elements $a, b \in S$ $f(a * b) = f(a) \, o \, f(b)$ is called a semi group homomorphism

A homomorphism of a semi group into itself is called a semi group endomorphism.

Identity element :- Let G be a non empty set and '$*$' be a binary operation on G. If there exists an element $e \in G$ such that $a * e = e * a = a$ for $a \in G$. Then 'e' is called an identity element on G.

Monoid :- A semi group $(G, *)$ with an identity element with respect to the binary operation '$*$' is known as a Monoid

Ex: $(Z, +)$ is a Monoid and the identity is '0'

$(Z, \cdot)$ is a Monoid and the identity is '1'.

Inverse element :- Let $(G, *)$ be an algebraic structure with the identity element 'e' in G w.r.t '$*$'. An element $a \in G$ is said to be invertible if there exist an element $x \in G$ such that $a * x = x * a = e$.

Group :- If G is a nonempty set and $*$ is a binary operation defined on G such that the following Laws are satisfied then $(G, *)$ is a group.

i) closure Law : Let $a, b \in G \Rightarrow a * b \in G$

ii) Associative Law : For $a, b, c \in G \Rightarrow a * (b * c) = (a * b) * c$

iii) Identity Law : For $a \in G$ there exist $e \in G$ such that $a * e = e * a = a$.

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iv) Inverse Law :- For $a \in G$ there exist an element $b \in G$ such that $a * b = b * a = e$, b is called an inverse of 'a'

Ex:- prove that $\{1, \omega, \omega^2\}$ is a group with respect to multiplication where $1, \omega, \omega^2$ are cube roots of unity.

Sol: we construct the composition table as follows.

$\cdot$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	$\omega^3=1$	ω

$\omega^3 = 1$, $\omega^4 = \omega^3 \cdot \omega = 1 \cdot \omega = \omega$

i) closure Law :- From the composition table it is clear that $(G, \cdot)$ is closed w.r.to multiplication.

ii) Associative Law :- From the composition table it is clear that $(G, \cdot)$ is Associative w.r.to multiplication.

iii) Identity Law :- '1' is identity element in G such that $a \cdot 1 = 1 \cdot a = 1$, $\forall a \in G$.

iv) Inverse Law :- Each element of G is invertible

$1 \cdot 1 = 1 \cdot 1 = 1 \Rightarrow$ '1' is its own inverse

$\omega \cdot \omega^2 = \omega^3 = 1 \Rightarrow \omega^2$ is the inverse of ω.

Hence $(G, \cdot)$ is a group.

Note :- $(G, \cdot)$ is a group and $a \cdot b = b \cdot a$, $\forall a, b \in G$ that is commutative Law holds in G with respect to the binary operation. then $(G, \cdot)$ is an abelian group.

EX:- show that the set $G = \{1, -1, i, -i\}$ where $i^2 = 1$ is an abelian group with respect multiplication as a binary operation.

Sol: we construct the composition table as follows

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·	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

i) Closure Law :- From the table $(G,\cdot)$ is closed w.r.to multiplication

ii) Associative Law :- For any three elements $a,b,c \in G \Rightarrow$

$a\cdot(b\cdot c) = (a\cdot b)\cdot c$

since $1\cdot(-1\cdot i) = 1\cdot -i = -i$

$(1\cdot -1)\, i = -1\cdot i = -i$

$\therefore$ Associative Law is satisfied.

iii) Identity Law :- '1' is the identity element in $(G,\cdot)$ such that $a\cdot 1 = 1\cdot a = a$, $\forall\, a \in G$.

iv) Inverse Law :- $1\cdot 1 = 1\cdot 1 = 1 \Rightarrow$ '1' is inverse of '1'

$(-1)\cdot(-1) = (-1)\cdot(-1) = 1 \Rightarrow$ '-1' is the inverse of '-1'

$i\cdot -i = -i\cdot i = 1 \Rightarrow$ 'i' is the inverse of '-i'

Hence Inverse of every element in G exists.

v) Commutative Law :- For $a,b \in G \Rightarrow a\cdot b = b\cdot a$

$1\cdot 1 = 1\cdot 1 = 1$, $-1\cdot 1 = 1\cdot -1 = -1$

$i\cdot 1 = 1\cdot i = i$, $-i\cdot i = i\cdot(-i) = 1$ etc...

Commutative Law is satisfied.

Hence $(G,\cdot)$ is an abelian group.

Ex: Prove that the set 'z' of all integers with binary operation '*' defined by $a*b = a+b+1$, $\forall\, a,b \in z$ is an abelian group.

Sol: Given $a*b = a+b+1$, $\forall\, a,b \in z$

i) closure Law :- Let $a,b \in z \Rightarrow a*b = a+b+1 \in z$

$\therefore$ 'z' is closed under '*'

ii) Associative Law :- Let $a,b,c \in z$

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$(a * b) * c = (a+b+1) * c$

$= a+b+1+c+1 = a+b+c+2$

$a * (b * c) = a * (b+c+1)$

$= a+b+c+1+1 = a+b+c+2$

$\therefore a * (b * c) = (a * b) * c.$

iii) Identity Law :- Let $a \in z$, there exist $e \in z$ such that $a * e = e * a = a$

$a+e+1 = a \Rightarrow e+1 = 0 \Rightarrow e = -1$

$\therefore e = -1$ is the identity element in Z.

iv) Inverse Law :- Let $a \in z$, there exist $b \in z$ such that $a * b = e$

$a+b+1 = -1 \Rightarrow a+b = -2 \Rightarrow b = -2-a$

$\therefore b = -2-a$ is an Inverse element in Z.

v) Commutative Law :- For $a, b \in G \Rightarrow a * b = b * a$

$a * b = a+b+1 = b+a+1 = b * a$

$\therefore$ Commutative Law is satisfied.

Hence $G, (Z, *)$ is an abelian group.

Ex: Show that the set Q_+ of all positive rational numbers forms an abelian group under the composition defined by $*$ such that $a * b = \frac{ab}{3}$ for $a, b \in Q_+$

Sol: Let $a, b \in Q_+ \Rightarrow a * b = \frac{ab}{3}$.

i) closure Law :- For $a, b \in Q_+ \Rightarrow a * b = \frac{ab}{3} \in Q_+$

ii) Associative Law : For $a, b, c \in Q_+$

$a * (b * c) = a * \frac{bc}{3} = \frac{\frac{abc}{3}}{3} = \frac{abc}{9}.$

$(a * b) * c = \frac{ab}{3} * c = \frac{\frac{ab}{3} \cdot c}{3} = \frac{abc}{9}$

$\therefore a * (b * c) = (a * b) * c,$ Associative Law satisfied.

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iii) Identity Law :- Let $a \in Q_+$ there exist $e \in Q_+$ such that $a \cdot e = e \cdot a = a$.

$$\frac{ae}{3} = a \Rightarrow ae - 3a = 0 \Rightarrow a(e-3) = 0 \quad (\because a \neq 0$$

$$\therefore e - 3 = 0 \Rightarrow \boxed{e = 3}$$

$\therefore e = 3$ is the identity element in Q_+

iv) Inverse Law :- Let $a \in Q_+$ there exist $b \in Q_+$ such that $a * b = e = b * a$

$$\frac{ab}{3} = e \Rightarrow \frac{ab}{3} = 3 \Rightarrow ab = 9 \Rightarrow \boxed{b = \frac{9}{a}}$$

$\therefore b = \frac{9}{a}$ is an inverse element in Q_+.

v) Commutative Law :- For $a, b \in Q_+ \Rightarrow a * b = b * a$

$$a * b = \frac{ab}{3} = \frac{ba}{3} = b * a.$$

$\therefore$ Commutative Law is satisfied.

Hence $(Q_+, *)$ is an abelian group.

EX: Prove that the set G of rational numbers other than '1' with operation $\oplus$ such that $a \oplus b = a + b - ab$ for $a, b \in G$ is abelian group. : practice problem

EX: Consider the algebraic system $(G, *)$ where 'G' is the set of all non-zero real numbers and '$*$' is a binary operation defined by $a * b = \frac{ab}{4}$ $\forall a, b \in G$. show that $(G, *)$ is an abelian group. : practice problem

Addition modulo 'm' :- If 'a' and 'b' are any two integers and 'r' is the least non-negative reminder obtaining by dividing the ordinary sum of 'a' and 'b' by 'm' then the addition modulo 'm' of 'a' and 'b' is 'r' symbolically $a +_m b = r$, $0 \leq r < m$

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EX: $20 +_6 5 = 1$ and $-15 +_5 3 = 3$

Multiplication modulo p: If 'a' and 'b' are any two integers and 'r' is the least non-negative reminder obtained by dividing the ordinary product of 'a' and 'b' by 'p' then the multiplication modulo 'p' of 'a' and 'b' is 'r' symbolically $a \times_p b = r$, $0 \le r < p$

EX: $5 \times_3 4 = 2$,

EX: show that the set $G = \{0,1,2,3,4\}$ is an abelian group with respect to addition modulo '5'.

sol: we form the composition table as follows.

$+_5$	0	1	2	3	4
0	[0]	1	2	3	4
1	1	2	3	4	[0]
2	2	3	4	[0]	1
3	3	4	[0]	1	2
4	4	[0]	1	2	3

i) closure Law :- since all the entries in the composition table are elements in G.

∴ closure Law satisfied.

ii) Associative Law :- For $a,b,c \in G$

$(a +_5 b) +_5 c = a +_5 (b +_5 c)$

$(1 +_5 3) +_5 4 = 1 +_5 (3 +_5 4) = 3$ etc.

iii) Identity Law :- clearly $0 \in G$ is the identity element

$0 +_5 a = 4 = a +_5 0$, $\forall a \in G$

iv) Inverse Law :- From the table

'0' is its own inverse ; '1' is the inverse of '4'

'2' is the inverse of 3 ; '3' is the inverse of '2'

'4' is the inverse of '1'.

v) commutative Law :- From the table

$a +_5 b = b +_5 a$, $\forall a,b \in G$

Hence $(G, +_5)$ is an abelian group.

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Ex: Show that the set $G = \{1, 2, 3, 4\}$ is an abelian group with respect to multiplication modulo 5.

Sol: The composition table for multiplication modulo '5' is

$\times_5$	1	2	3	4
1	[1]	2	3	4
2	2	4	[1]	3
3	3	[1]	4	2
4	4	3	2	[1]

i) **closure Law :-** since all the entries in the composition table are elements in G.

$\therefore$ closure Law is satisfied.

ii) **Associative Law :-** For $a, b, c \in G$

$(a \times_5 b) \times_5 c = a \times_5 (b \times_5 c)$

$(1 \times_5 3) \times_5 4 = 1 \times_5 (3 \times_5 4) = 2$ etc.

iii) **Identity Law :-** clearly $1 \in G$ is the identity element

$1 \times_5 a = a = a \times_5 1 \quad \forall a \in G$

iv) **Inverse Law :-** From the table

'1' is the inverse of '1'; '2' is the inverse of '3'

'3' is the inverse of '2'; '4' is the inverse of '4'

v) **Commutative Law :-** From the table

$a \times_5 b = b \times_5 a, \quad \forall a, b \in G$

Hence $(G, \times_5)$ is an abelian group.

Ex: Show that $G = \{x / x = 2^a 3^b \text{ for } a, b \in z\}$ is a group under multiplication.

Sol: Let $x, y, z \in G$, we can take

$x = 2^p 3^q, \quad y = 2^r 3^s, \quad z = 2^l 3^m, \quad p, q, r, s, l, m \in z$

i) **closure Law :-** For $x, y \in G \Rightarrow x \cdot y \in G$

since $x \cdot y = (2^p \cdot 3^q) \cdot (2^r \cdot 3^s)$

$= 2^{p+r} \cdot 3^{q+s} \in G.$

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ii) Associative Law :- For $x, y, z \in G$

$$x \cdot (y \cdot z) = x \cdot (y z)$$
$$= 2^p 3^q \left(2^r \cdot 3^s \cdot 2^l 3^m \right)$$
$$= 2^p 3^q \left(2^{r+l} 3^{s+m} \right)$$
$$= 2^{p+r+l} \cdot 3^{q+s+m}$$
$$= \left(2^{p+r} \cdot 3^{q+s}\right) 2^l \cdot 3^m$$
$$= \left(2^p \cdot 3^q \cdot 2^r \cdot 3^s\right) 2^l \cdot 3^m = (x \cdot y) \cdot z$$

$\therefore$ Associative is satisfied.

iii) Identity Law : Let $x \in G$

We know that $e = 2^0 3^0 \in G$, since $0 \in Z$

$$\therefore x \cdot e = 2^p 3^q \cdot 2^0 \cdot 3^0 = 2^{p+0} 3^{q+0} = 2^p 3^q = x$$

$\therefore x \cdot e = e \cdot x = x$

iv) Inverse Law :- Let $x \in G$

Now $y = 2^{-p} 3^{-q} \in G$, since $-p, -q \in Z$

$$\therefore x \cdot y = 2^p \cdot 3^q \cdot 2^{-p} \cdot 3^{-q} = 2^{p-p} 3^{q-q} = 2^0 \cdot 3^0 = e$$

$\therefore x \cdot y = y \cdot x = e$

Hence $(G, \cdot)$ is a group.

Ex: If $(G, *)$ is a group then $(a*b)^{-1} = b^{-1} * a^{-1}$, $\forall a, b \in G$

Sol: Let $a, b \in G$ and 'e' is the identity element in G

Let $a \in G \Rightarrow a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$

$b \in G \Rightarrow b^{-1} \in G$ such that $b * b^{-1} = b^{-1} * b = e$

Now $a, b \in G \Rightarrow a * b \in G$ and $(a*b)^{-1} \in G$

Consider

$$(a*b) * (b^{-1} * a^{-1}) = a * [\, b * (b^{-1} * a^{-1}) \,]$$
$$= a * [\, (b * b^{-1}) * a^{-1} \,]$$

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$$= a * (e * a^{-1})$$

$$= a * a^{-1} = e$$

$$(b^{-1} * a^{-1}) * (a * b) = b^{-1} * [a^{-1} * (a * b)]$$

$$= b^{-1} * [(a^{-1} * a) * b]$$

$$= b^{-1} * (e * b)$$

$$= b^{-1} * b = e$$

$$\therefore (a * b) * (b^{-1} * a^{-1}) = (b^{-1} * a^{-1}) * (a * b) = e$$

Hence $(a * b)^{-1} = b^{-1} * a^{-1}$

Ex: Cancelation Laws holds in G

For all $a, b, c \in G$ then $a * b = a * c \Rightarrow b = c$ (L.C.L)

$b * a = c * a \Rightarrow b = c$ (R.C.L)

Sol: G is a group

Let 'e' be the identity element in G.

$a \in G \Rightarrow a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$

Consider $a * b = a * c$

$\Rightarrow a^{-1} * (a * b) = a^{-1} * (a * c)$

$\Rightarrow (a^{-1} * a) * b = (a^{-1} * a) * c$

$\Rightarrow e * b = e * c \Rightarrow b = c$

Now $b * a = c * a$

$\Rightarrow (b * a) * a^{-1} = (c * a) * a^{-1}$

$\Rightarrow b * (a * a^{-1}) = c * (a * a^{-1})$

$\Rightarrow b * e = c * e \Rightarrow b = c$ //

Ex: If every element of a group G is its own inverse show that G is an abelian group.

Sol: Let $a, b \in G$ By Hypothesis $a^{-1} = a$, $b^{-1} = b$

Then $ab \in G \Rightarrow (ab)^{-1} = ab$

Now $(ab)^{-1} = ab \Rightarrow b^{-1}a^{-1} = ab \Rightarrow ba = ab$

$\therefore$ 'G' is an abelian group.

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Ex: Prove that if $a^2 = a$ then $a = e$, 'a' being an element of a group G.

Sol: Let 'a' be an element of a group G such that $a^2 = a$

Now $a^2 = a \Rightarrow aa = a$

$\Rightarrow (aa)a^{-1} = aa^{-1}$

$\Rightarrow a(aa^{-1}) = aa^{-1}$ $\quad \langle \because aa^{-1} = e$

$\Rightarrow ae = e \Rightarrow \boxed{a = e}$

Ex: Show that in a group G, for $a, b \in G$, $(ab)^2 = a^2b^2$ $\Leftrightarrow$ G is an abelian.

Sol: Let $a, b \in G$ and $(ab)^2 = a^2b^2$ Then

$(ab)^2 = a^2b^2 \Rightarrow (ab)(ab) = a^2b^2 = (aa)(bb)$

$\Rightarrow a(ba)b = a(ab)b$

$\Rightarrow ba = ab$ (by cancellation Law)

$\Rightarrow$ 'G' is abelian

Conversely, Let G is an abelian Then

$(ab)^2 = (ab)(ab) = a(ba)b = a(ab)b = (aa)(bb) = a^2b^2$ //.

Ex: If a, b are any two elements of a group $(G, \cdot)$ which commute. Show that 1. a^{-1} and 'b' commute 2. b^{-1} and 'a' commute 3. 'a^{-1}' and 'b^{-1}' commute

Sol: $(G, \cdot)$ is a group such that $ab = ba$

1. $ab = ba \Rightarrow a^{-1}(ab) = a^{-1}(ba)$

$\Rightarrow (a^{-1}a)b = a^{-1}(ba)$

$\Rightarrow eb = (a^{-1}b)a \Rightarrow b = (a^{-1}b)a$

$\Rightarrow ba^{-1} = (a^{-1}b)aa^{-1}$

$\Rightarrow ba^{-1} = (a^{-1}b)e$

$\Rightarrow ba^{-1} = a^{-1}b \Rightarrow$ 'a^{-1}' and 'b' commute.

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2. $ab = ba \Rightarrow (ab)\,b^{-1} = (ba)\,b^{-1}$

$\Rightarrow a(bb^{-1}) = (ba)\,b^{-1}$

$\Rightarrow ae = b(ab^{-1})$

$\Rightarrow a = b(ab^{-1})$

$\Rightarrow b^{-1}a = b^{-1}(b(ab^{-1}))$

$\Rightarrow b^{-1}a = b^{-1}b\,(ab^{-1}) \Rightarrow b^{-1}a = e(ab^{-1})$

$\Rightarrow b^{-1}a = ab^{-1}$

$\therefore$ b^{-1} and 'a' commute

3. $ab = ba \Rightarrow (ab)^{-1} = (ba)^{-1} \Rightarrow b^{-1}a^{-1} = a^{-1}b^{-1}$

$\therefore$ 'a^{-1}' and 'b^{-1}' are commute.

order of an element :- Let $(G, *)$ be a group and $a \in G$ then the least positive integer 'n' if it exist such that $a^n = e$ is called the order of $a \in G$. The order of an element $a \in G$ is be denoted by $o(a)$.

Ex: If $G = \{1, -1, i, -i\}$

$1^1 = 1^2 = 1^3 = \dots = 1 \Rightarrow o(1) = 1$

$(-1)^2 = (-1)^4 = (-1)^6 = \dots = 1 \Rightarrow o(-1) = 2$

$i^4 = i^8 = \dots = 1 \Rightarrow o(i) = 4$

$(-i)^4 = (-i)^8 = \dots = 1 \Rightarrow o(-i) = 4.$

Sub groups :- Let $(G, *)$ be a group and 'H' be a non empty subset of G. If $(H, *)$ is itself is a group then $(H, *)$ is called sub-group of $(G, *)$

Ex: i) $(Z, +)$ is a sub group of $(Q, +)$

ii) $(N, +)$ is not a sub group of the group $(Z, +)$.

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Ex: If $(G,*)$ is a group and $H \subseteq G$ then $(H,*)$ is a subgroup of $(G,*)$ if and only if i) $a,b \in H \Rightarrow a*b \in H$

ii) $a \in H \Rightarrow a^{-1} \in H$.

Sol: Let $(H,*)$ be a sub group of $(G,*)$

since $(H,*)$ is a group

By closure property we have $a,b \in H \Rightarrow ab \in H$

By inverse property $a \in H \Rightarrow a^{-1} \in H$.

Conversely Let $a,b \in H \Rightarrow ab \in H$ and $a \in H \Rightarrow a^{-1} \in H$.

since H is non empty

Let $a \in H \Rightarrow a^{-1} \in H$

$\therefore a \in H,\ a^{-1} \in H \Rightarrow aa^{-1} \in H \Rightarrow e \in H$, e is identity in G

$\Rightarrow$ 'e' is the identity in H.

'*' is associative in H follows from the fact that '*' is associative in G.

Hence H Itself is a group, $\therefore$ H is a subgroup of 'G'.

Ex: If H_1 and H_2 are two subgroups of a group G. then $H_1 \cap H_2$ is also a sub group of G.

Sol: Let H_1 and H_2 be two subgroups of a group G.

Let 'e' be the identity element in G.

$\therefore e \in H_1$ and $e \in H_2 \Rightarrow e \in H_1 \cap H_2 \Rightarrow H_1 \cap H_2 \neq \phi$

Let $a \in H_1 \cap H_2$, and $b \in H_1 \cap H_2$

$a \in H_1, a \in H_2$ and $b \in H_1, b \in H_2$

since H_1 is a sub group, $a \in H_1$ and $b \in H_1 \Rightarrow ab^{-1} \in H_1$

similarly $ab^{-1} \in H_2$ $\therefore ab^{-1} \in H_1 \cap H_2$

we have $a \in H_1 \cap H_2$, $b \in H_1 \cap H_2 \Rightarrow ab^{-1} \in H_1 \cap H_2$

$\therefore H_1 \cap H_2$ is a subgroup of G.

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